DIY_SPOT_WELDER_V1 — SYSTEM OVERVIEW (Architecture C: MOSFET pulse switch + carry-only isolation relay + fuse + manual disconnect)

```
BT1 RT1270 (+) --F101 ANL--S101 DISCONNECT---+ P_BUS --K101 relay (carry-only)-- ELECTRODE+ ~~ workpiece ~~ ELECTRODE- --+ DRAIN_BUS
                                             |                                                                        |
                                             +--F102 3A--> J1 control board (hold-up banks, 7805, Nano, interlocks)   6x IRLZ44N
                                             +--D101 STPS3045CP (free-wheel, cathode)  <---------- anodes ---------- DRAIN_BUS
BT1 (-) ==16 mm2== SOURCE bar = GND star point <-- control GND (J1.2, J3.2/J3.6) ---------------------------------- sources
```

Safety layers that do NOT depend on the MCU: F101 fuse | S101 disconnect | S102 E-STOP 2NC | S103 ARM 2NO | R45 default-off pull-down |
U3A pulse-length limiter (63-96 ms) | U3B gate-drive UVLO (8 V) | K101 opens when ARM chain opens.
Firmware layers: weld-enable state machine, latched faults, watchdog, stuck-on detection (battery sag / drain), thermal derating.

STATUS: PRELIMINARY. The welding source (RT1270) is NOT approved (estimated 200-390 A, 4x its datasheet max). K101 ratings unknown.

The control PCB contains no welding-current conductors. No manufacturing outputs are released. See OPEN_ISSUES.md / RELEASE_CHECKLIST.md.

02_POWER_INPUT_AND_PROTECTION

Power input, protection, hold-up and regulators

03_CONTROL_AND_TIMING

Microcontroller (Arduino Nano) and timing

04_SWITCHING_AND_GATE_DRIVE

Gate driver and MOSFET power assembly (assembly = off-board)

05_RELAY_CONTROL_AND_ISOLATION

Isolation relay, main fuse, disconnect, ARM / E-STOP chain

06_CURRENT_VOLTAGE_SENSING

Voltage sensing (battery, drain, electrode, rails)

07_OLED_AND_USER_INTERFACE

OLED, encoder, buttons, buzzer, LEDs

08_TEMPERATURE_MONITORING

Heatsink NTC and board temperature sensor

09_TRIGGER_AND_INTERLOCKS

Trigger conditioning, hardware pulse limiter, gate-drive UVLO

10_SERVICE_AND_TEST_POINTS

Test points and mechanical

Derived concepts from Arduino Spot Welder V4 (KaeptnBalu / Malectrics, GPL-3.0)
Sheet 01 of 10
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DIY_SPOT_WELDER_V1 project — GPL-3.0-or-later

Sheet: /
File: DIY_SPOT_WELDER_V1.kicad_sch

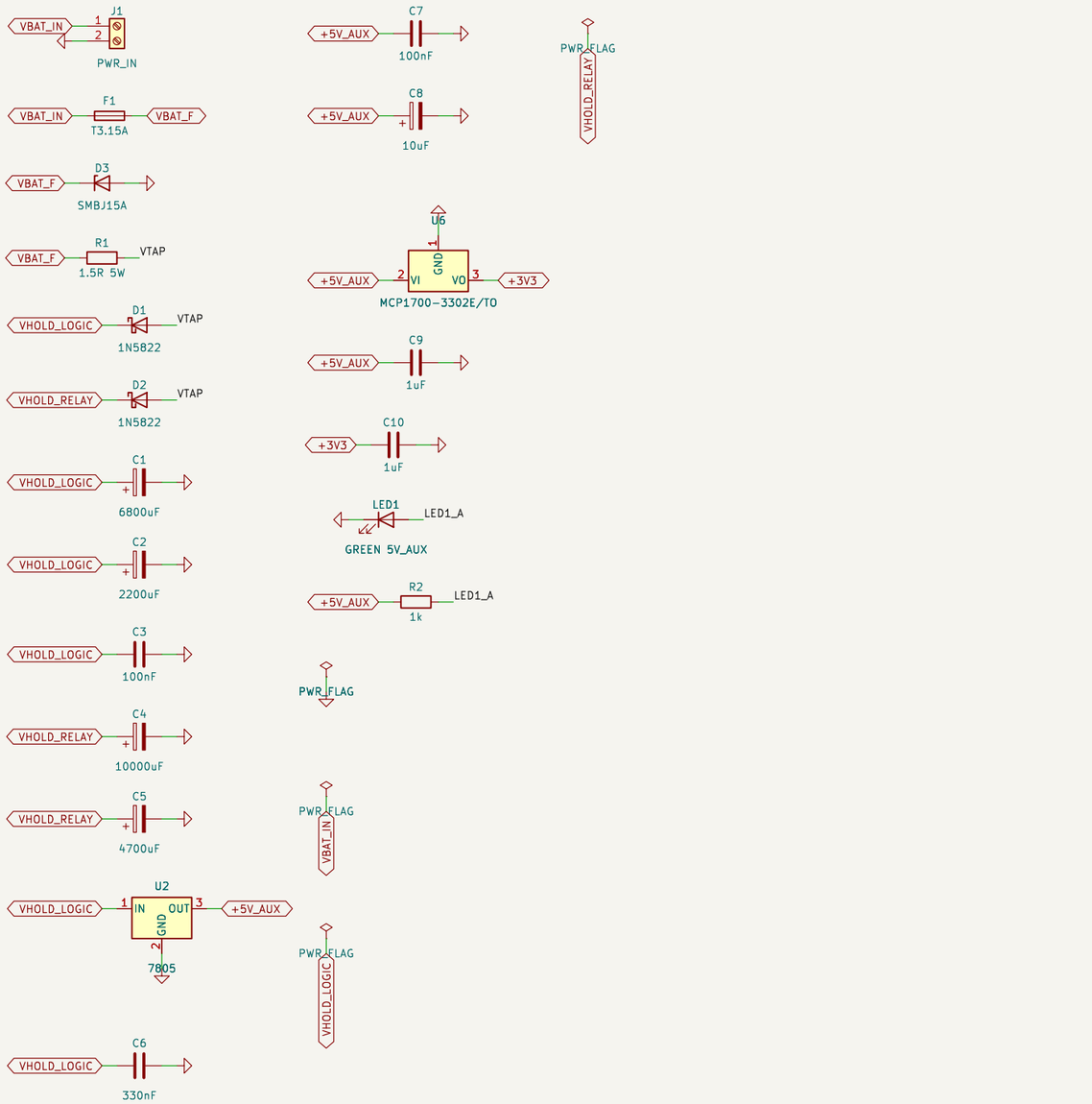
Title: DIY_SPOT_WELDER_V1 — System overview

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10

POWER INPUT: fused tap from P_BUS (F102 3 A at source + F1 T3.15A on board), TVS D3, inrush/limit R1,
two diode-isolated hold-up banks: VHOLD_LOGIC (C1+C2 = 9 mF) and VHOLD_RELAY (C4+C5 = 14.7 mF).

Reason: the RT1270 terminal voltage collapses to ~3 V during a weld (doc 03 section 3) — rails must not follow it.

U2 7805 -> +5V_AUX (hardware references, HCT14, LEDs, buzzer). U6 MCP1700 -> +3V3 (OLED, MCP9804, I2C).



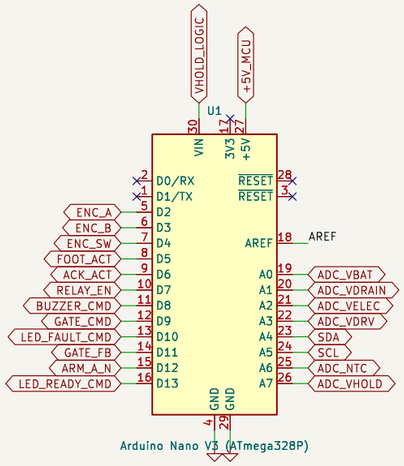
Derived concepts from Arduino Spot Welder V4 (KaeptnBalu / Malectrics, GPL-3.0)
Sheet 02 of 10
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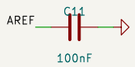
Sheet: /02_POWER_INPUT_AND_PROTECTION/
File: 02_POWER_INPUT_AND_PROTECTION.kicad_sch

Title: DIY_SPOT_WELDER_V1 — Power input, protection, hold-up and regulator

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/10

MCU: Arduino Nano V3 in sockets. VIN <- VHOLD_LOGIC. Nano 5 V output (+5V_MCU) = ADC reference + NTC pull-up.
D9 = GATE_CMD (Timer1 OC1A capable). All outputs are inputs (hi-Z) during reset/bootloader -> hardware pull-downs keep
the weld OFF. The MCU is NOT the only means of ending a pulse (see sheet 09).







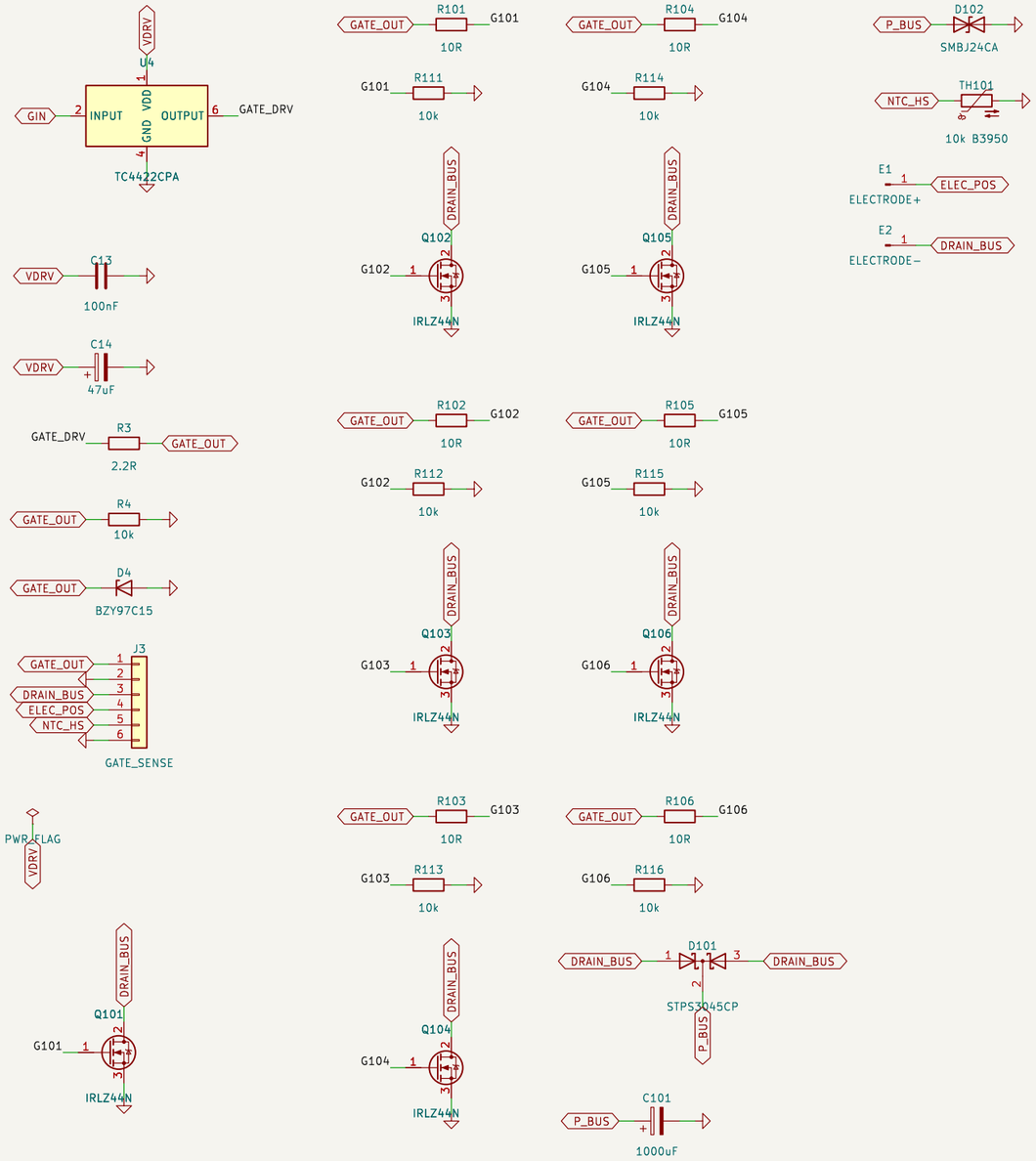
Derived concepts from Arduino Spot Welder V4 (KaeptnBalu / Malectrics, GPL-3.0)
Sheet 03 of 10
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DIY_SPOT_WELDER_V1 project - GPL-3.0-or-later

Sheet: /03_CONTROL_AND_TIMING/
File: 03_CONTROL_AND_TIMING.kicad_sch

Title: DIY_SPOT_WELDER_V1 - Microcontroller (Arduino Nano) and timing

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/10

GATE DRIVE: TC4422 (non-inverting) powered from VDRV (only present when E-STOP closed AND ARM on).
POWER ASSEMBLY (off-board, 'exclude from board'): 6x IRLZ44N on copper busbars, 10R gate + 10k G-S per device,
STPS3045CP free-wheel diode drain->P_BUS, 1000 uF + SMBJ24CA across P_BUS-SOURCE. Welding current NEVER flows on the PCB.
Rating limited by MOSFET thermal stress: <=50 ms at ~300 A (doc 03 section 4). MOSFETs can FAIL SHORT: see sheet 05 + doc 04.



Derived concepts from Arduino Spot Welder V4 (KaeptnBalu / Malectrics, GPL-3.0)
Sheet 04 of 10
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DIY_SPOT_WELDER_V1 project - GPL-3.0-or-later

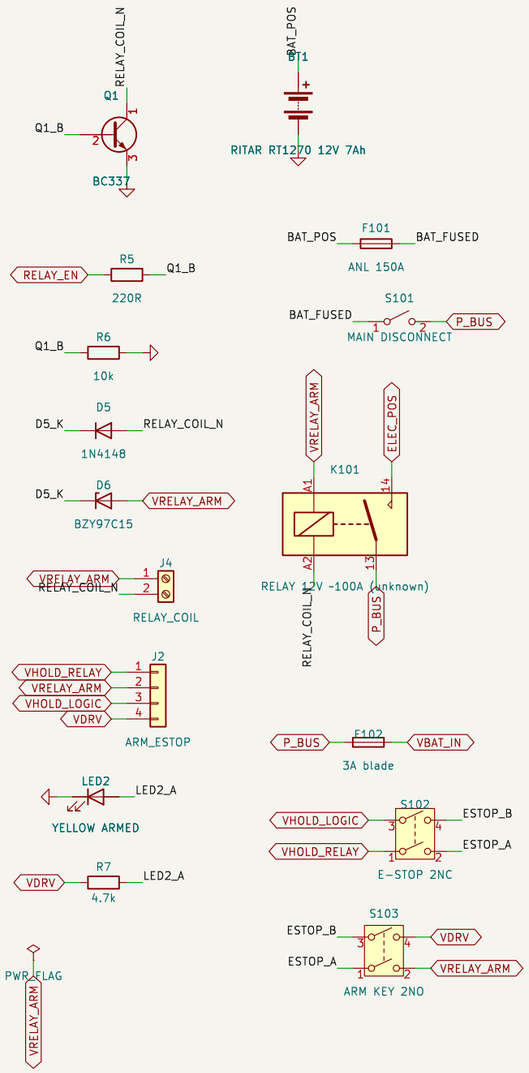
Sheet: /04_SWITCHING_AND_GATE_DRIVE/
File: 04_SWITCHING_AND_GATE_DRIVE.kicad_sch

Title: DIY_SPOT_WELDER_V1 - Gate driver and MOSFET power assembly (asser

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/10

ISOLATION: K101 relay is closed before welding and opened only when no current flows (carry-only).
Two-pole E-STOP (S102) and ARM (S103) chains remove relay-coil supply (pole A) and gate-driver supply (pole B) in hardware.

F101 main fuse + S101 manual disconnect are the backstop for a shorted MOSFET. K101 ratings are UNKNOWN — provisional.



Derived concepts from Arduino Spot Welder V4 (KaeptnBalu / Malectrics, GPL-3.0)
Sheet 05 of 10
Generated by KiCad/Generator/gen_schematic.py from design.py (single source)
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DIY_SPOT_WELDER_V1 project — GPL-3.0-or-later

Sheet: /05_RELAY_CONTROL_AND_ISOLATION/
File: 05_RELAY_CONTROL_AND_ISOLATION.kicad_sch

Title: DIY_SPOT_WELDER_V1 — Isolation relay, main fuse, disconnect, ARM / E

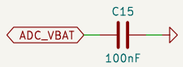
Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/10

SENSING: five 33k/10k dividers (k = 4.30), 100 nF filters. During a pulse the firmware switches to the 1.1 V
reference (4.6 mV/LSB at the input of the divider ~ 20 mV) to estimate V_DS and the battery sag.

Current = V_DS / RDS(on) estimate (+−30 %), shown as 'EST.' only. No shunt in V1 (upgrade path in OPEN_ISSUES).









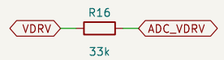


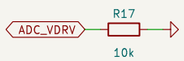


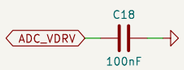


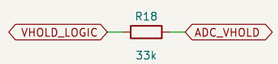


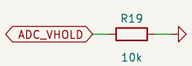














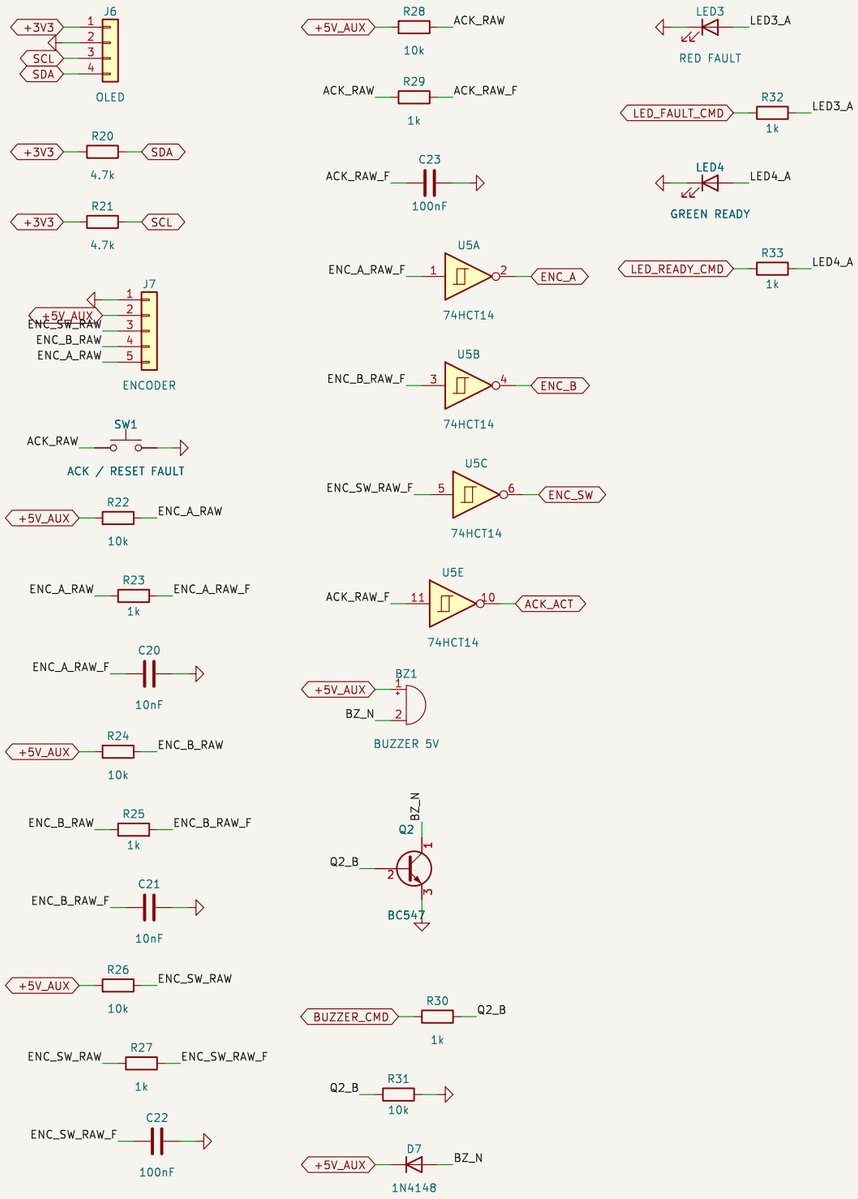
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Sheet 06 of 10
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Sheet: /06_CURRENT_VOLTAGE_SENSING/
File: 06_CURRENT_VOLTAGE_SENSING.kicad_sch

Title: DIY_SPOT_WELDER_V1 − Voltage sensing (battery, drain, electrode, rails

Size: A3	Date: 2026−10−12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 6/10

UI: OLED on 3.3 V I2C (pull-ups to 3.3 V), EC11 encoder and ACK button through 74HCT14 Schmitt triggers with RC filters.
Active buzzer via BC547, LEDs: FAULT (red), READY (green), ARMED (yellow, hardware on VDRV), 5V_AUX (green).



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Sheet 07 of 10
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DIY_SPOT_WELDER_V1 project — GPL-3.0-or-later

Sheet: /07_OLED_AND_USER_INTERFACE/
File: 07_OLED_AND_USER_INTERFACE.kicad_sch

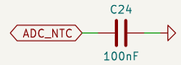
Title: DIY_SPOT_WELDER_V1 — OLED, encoder, buttons, buzzer, LEDs

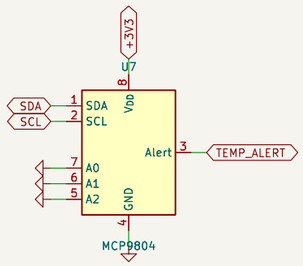
Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/10

TEMPERATURE: TH101 NTC on the drain bar (insulated) -> A6 (ratiometric). MCP9804 on board (I2C 0x18) for ambient/board.
Firmware derates the maximum pulse with heatsink temperature and locks out at 60 C (latched).

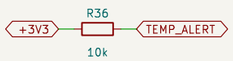












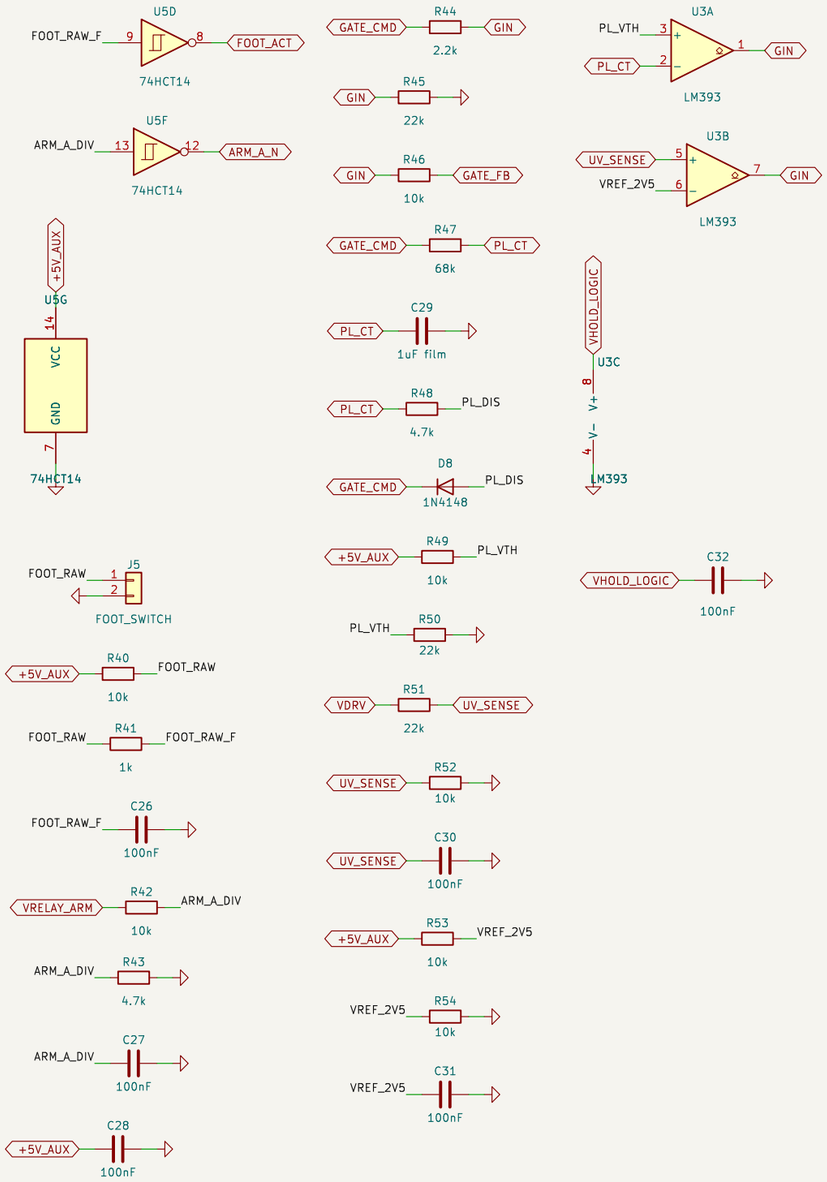
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DIY_SPOT_WELDER_V1 project - GPL-3.0-or-later

Sheet: /08_TEMPERATURE_MONITORING/
File: 08_TEMPERATURE_MONITORING.kicad_sch

Title: DIY_SPOT_WELDER_V1 - Heatsink NTC and board temperature sensor

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/10

INTERLOCKS (independent of the MCU): GIN is a wired-AND node.
U3A: pulse-length limiter. PL_CT charges from GATE_CMD through 68k/1uF; when it exceeds 0.6875 x 5 V the LM393 pulls GIN low.
Window 63-96 ms (Monte-Carlo, doc 03 sec. 8); firmware limit 50 ms. Reset through D8/R48 when GATE_CMD returns low.
U3B: gate-drive UVLO. If VDRV < 8.0 V the LM393 pulls GIN low (prevents linear-mode operation of the MOSFETs).
R45 22k pull-down: GIN low when the MCU is reset or unpowered. GATE_FB lets the firmware verify the effective command.
Foot switch: pull-up + RC + Schmitt + firmware re-confirmation for the whole trigger delay (fixes reference finding).



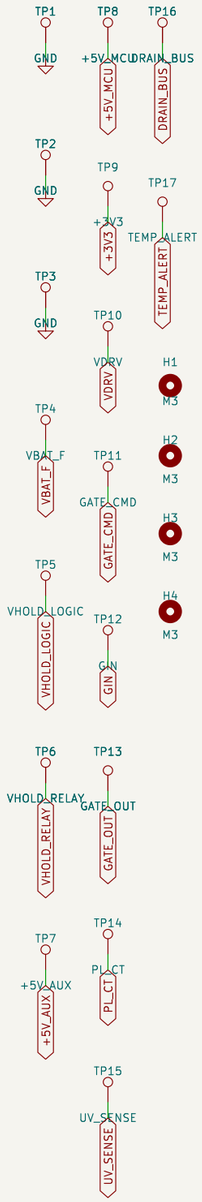
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Sheet 09 of 10
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DIY_SPOT_WELDER_V1 project - GPL-3.0-or-later

Sheet: /09_TRIGGER_AND_INTERLOCKS/
File: 09_TRIGGER_AND_INTERLOCKS.kicad_sch

Title: DIY_SPOT_WELDER_V1 - Trigger conditioning, hardware pulse limiter, ga

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/10

TEST POINTS for bench validation without the welding stage connected (see RELEASE_CHECKLIST.md).
Mounting holes are isolated; use nylon spacers.



Derived concepts from Arduino Spot Welder V4 (KaeptnBalu / Malectrics, GPL-3.0)
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Sheet: /10_SERVICE_AND_TEST_POINTS/
File: 10_SERVICE_AND_TEST_POINTS.kicad_sch

Title: DIY_SPOT_WELDER_V1 — Test points and mechanical

Size: A3	Date: 2026-10-12	**Rev: A (PRELIMINARY)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/10